Hw9

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Question 1

The complete candidate settings and their mean five-fold cross-validation misclassification rates are reported in the output table. I prefer setting 7, the degree-3 polynomial SVM with cost $C = 0.1$, fixed polynomial $\gamma = 1/9$, and `coef0 = 1`. It has the lowest mean CV error among the 18 candidates (0.2598878). The next best candidate is the degree-2 polynomial SVM with $C = 1$ (0.2684432), while the best radial candidate has $C = 10$ and $\gamma = 0.01$ (0.2707106). Gamma does not apply to the linear kernel, so its `NA` entries are expected and do not prevent comparison. A smaller C penalizes margin violations less strongly; its value alone does not establish better prediction. These CV results support the degree-3 polynomial choice among the settings tested, although the leading errors are close.

```
heart <- read.csv("/Users/zhijinghuang/Downloads/Stat432_2026/week-09/data/saheart.csv")

heart$famhist <- as.numeric(factor(heart$famhist, levels = c("Absent", "Present")))

X <- as.matrix(heart[, names(heart) != "chd"])
y <- factor(heart$chd, levels = c(0, 1))
```

```
c_values <- c(0.1, 1, 10)
settings <- rbind(
  data.frame(
    kernel = rep(c("linear", "poly2", "poly3", "poly4"), each = 3),
    C = rep(c_values, 4),
    gamma = NA_real_
  ),
  data.frame(
    kernel = "radial",
    C = rep(c_values, 2),
    gamma = rep(c(0.01, 0.1), each = 3)
  )
)
settings$gamma[startsWith(settings$kernel, "poly")] <- 1 / ncol(X)
set.seed(43209)
fold <- integer(nrow(heart))
for (class in levels(y)) {
  id <- which(y == class)
  fold[id] <- sample(rep(1:5, length.out = length(id)))
}
```

```r
errors <- matrix(NA_real_, nrow = nrow(settings), ncol = 5)

for (j in 1:5) {
  fit_id <- which(fold != j)
  valid_id <- which(fold == j)

  x_fit <- scale(X[fit_id, , drop = FALSE])
  x_valid <- scale(
    X[valid_id, , drop = FALSE],
    center = attr(x_fit, "scaled:center"),
    scale = attr(x_fit, "scaled:scale")
  )

  for (i in seq_len(nrow(settings))) {
    name <- settings$kernel[i]

    kernel <- if (startsWith(name, "poly")) "polynomial" else name

    degree <- if (startsWith(name, "poly")) {
      as.integer(sub("poly", "", name))
    } else {
      3
    }

    gamma <- if (name == "radial") {
      settings$gamma[i]
    } else {
      1 / ncol(X)
    }

    fit <- e1071::svm(
      x = x_fit,
      y = y[fit_id],
      type = "C-classification",
      kernel = kernel,
      cost = settings$C[i],
      degree = degree,
      gamma = gamma,
      coef0 = 1,
      scale = FALSE
    )

    predicted <- predict(fit, x_valid)
    errors[i, j] <- mean(predicted != y[valid_id])
  }
}
```

```r
settings$mean_cv_error <- rowMeans(errors)
settings[order(settings$mean_cv_error), ]
```

```
##    kernel    C     gamma mean_cv_error
## 7   poly3  0.1 0.1111111     0.2598878
## 5   poly2  1.0 0.1111111     0.2684432
```

```
## 15 radial 10.0 0.0100000     0.2707106
## 4   poly2  0.1 0.1111111     0.2750117
## 1  linear  0.1        NA     0.2772557
## 17 radial  1.0 0.1000000     0.2793829
## 10  poly4  0.1 0.1111111     0.2794063
## 14 radial  1.0 0.0100000     0.2815334
## 6   poly2 10.0 0.1111111     0.2815802
## 3  linear 10.0        NA     0.2836840
## 2  linear  1.0        NA     0.2880318
## 18 radial 10.0 0.1000000     0.3139084
## 8   poly3  1.0 0.1111111     0.3140019
## 11  poly4  1.0 0.1111111     0.3269051
## 9   poly3 10.0 0.1111111     0.3333100
## 13 radial  0.1 0.0100000     0.3463301
## 16 radial  0.1 0.1000000     0.3463301
## 12  poly4 10.0 0.1111111     0.3571061
```

QUESTION 2 - ONE-PAGE SKILL REPORT

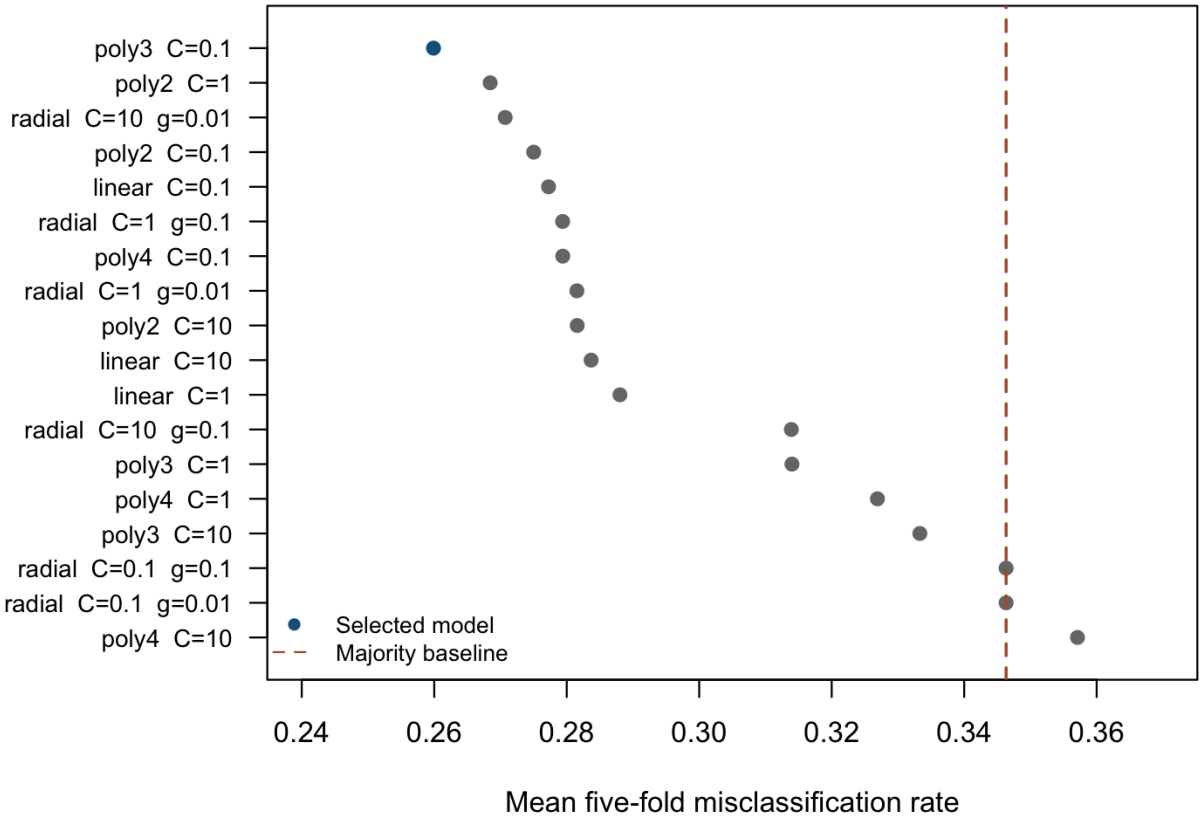
SVM analysis of SAheart

Suitability: 462 rows, nine numeric predictors after famhist recoding, no missing values, and a binary chd outcome (160/462 = 34.6% positive). Median age is 45; median sbp is 134. A majority-class predictor has 0.3463 mean five-fold error.

Evaluation: linear tests a straight boundary; polynomial degrees 2-4 and radial allow nonlinear boundaries. Compare 18 settings on the same five stratified folds with fitting-only scaling. C (margin-violation cost) is 0.1, 1, or 10; radial gamma (locality) is 0.01 or 0.1. Polynomial gamma = 1/9 and coef0 = 1. Select by mean validation error.

Best three candidates, ordered by mean CV error

Kernel	C	Gamma	Mean error
poly3	**0.1**	**1/9**	**0.2599**
poly2	1	1/9	0.2684
radial	10	0.01	0.2707



Selected: degree-3 polynomial, C = 0.1; mean CV error 0.2599, or 8.64 percentage points below the majority baseline. The best radial setting reaches 0.2707. Across out-of-fold predictions, 279 negatives and 63 positives were correct; 23 negatives and 97 positives were wrong. Sensitivity for chd = 1 is 39.4%, so the model misses many positives. The chosen settings were refit on all 462 rows for future prediction.

APPENDIX

Complete hand-written SKILL.md

```
---
name: svm-analysis
description: analyze svm, and pick the final model from it.
---

# svm analysis
-   this model is written on purpose to evaluate SVM analysis, and choose the best
model to be the final model based on the analyzation.

-   read the provided data, match the data with the requirements for a SVM analysis.
make sure whether the CV error is evaluated through a simple baseline.
If not, base the baseline as predicting the majority(> 70%) of the data. report
data dimensions, type, response variables, predictors and summary statistics.

-   sort the outputs from small to large by the mean cv error.
A preferred best model usually comes with the one with the smallest cv error,
but check other parameters to confirm they are not causing enormous effect to
the result. After that, provide the reason for your selection.

-   check for tuning parameters, such as C values and kernel scales, and report
them with data analysis. apply the same folds for every candidate.

-   for the output, summarize the statistical results in a table, ranking from
small to large; if the results can be visualized, visualize them in a 6.5 by 4.5
inch figure. make sure the table and the figure allign with the results, and can
be seen as justification to the choices made.

-   in the output, interpret the fit by interpreting tuning parameters, the
response, the mean CV error, and anything else that will significantly affect
the model.

-   list the evidence for your model choice, and prioritize the evidence needed
to justify the final model, along with the output table and graphs, in no more
than one printed-page at a 11-point font.
```